

DropBox :

Functional requirements:

user should be able to upload/download a file based on his preference,
user should be able to search for any file based on his preference,
user should be able to upload any file more than 50GB

Non-functional requirements:

Data-partitioning,

load balancing,

fault tolerance,

vertical scalability,

horizontal scalability,

low latency,

high throughput,

CAP theorem (Here eventual consistency is fine , but here we need 99.99% uptime

i.e Availability>>Consistency)

Core Entities:

user ,

file,

history,

API :

POST:/upload

```
{
  userID,
  {fileMetadata}
}
```

GET:/search/{fileID}

